

PE-EC701A — Microwave Theory and Technique

Semester VII • 3L:0T:0P • 3 Credits

Module 3 Exam Notes: RF and Microwave Transmission Lines

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

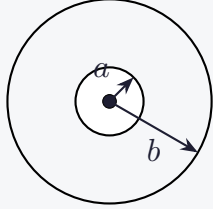
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1. Coaxial Line

The coaxial line is a standard TEM transmission line with two concentric circular conductors. It operates from DC up to frequencies where higher-order waveguide modes (usually TE_{11}) begin to propagate.

Coaxial Line Structure



Conductors:

Inner Radius = a

Outer Radius = b

Dielectric constant = ϵ_r

Coaxial Line Characteristic Impedance

The characteristic impedance Z_0 of a lossless coaxial line is given by:

$$Z_0 = \frac{\eta_0}{2\pi\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right) \approx \frac{138}{\sqrt{\epsilon_r}} \log_{10}\left(\frac{b}{a}\right) \quad (\Omega) \quad (1)$$

Where:

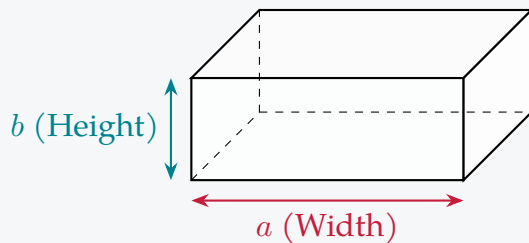
$\eta_0 \approx 377 \Omega$ is the intrinsic impedance of free space.

ϵ_r is the relative permittivity of the dielectric.

2. Rectangular Waveguide

A rectangular waveguide is a hollow metallic pipe of rectangular cross-section. Since it has only one conductor, it cannot support TEM waves; propagation occurs via TE and TM modes.

Rectangular Waveguide Geometry



Dominant dimension relation:

$$a > b$$

Typically, $a = 2b$ to maximize single-mode bandwidth.

2.1 Wave Equations and Mode Cutoff

From Helmholtz wave equations inside a lossless rectangular guide:

$$f_c = \frac{v_0}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{c}{2\sqrt{\epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \quad (2)$$

Where m and n are mode integers. The **dominant mode** (lowest cutoff frequency) for $a > b$ is the **TE₁₀ mode** ($m = 1, n = 0$):

$$f_{c(\text{TE}_{10})} = \frac{c}{2a\sqrt{\epsilon_r}} \quad (3)$$

2.2 Phase Velocity, Group Velocity, and Guide Wavelength

Waveguide propagation constants vary non-linearly with frequency:

- **Guide Wavelength (λ_g):** The distance between two equal-phase planes along the guide axis.

$$\lambda_g = \frac{\lambda_0}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} > \lambda_0 \quad (4)$$

- **Phase Velocity (v_p):** The velocity at which the wave's phase fronts propagate.

$$v_p = \frac{\omega}{\beta} = \frac{v_0}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} > v_0 \quad (5)$$

- **Group Velocity (v_g):** The velocity at which real energy or information travels.

$$v_g = \frac{d\omega}{d\beta} = v_0 \sqrt{1 - \left(\frac{f_c}{f}\right)^2} < v_0 \quad (6)$$

2.2.1 Derivation of the Velocity Relation

We multiply the expressions for phase and group velocity:

$$v_p \cdot v_g = \left(\frac{v_0}{\sqrt{1 - (f_c/f)^2}} \right) \times \left(v_0 \sqrt{1 - (f_c/f)^2} \right) = v_0^2$$

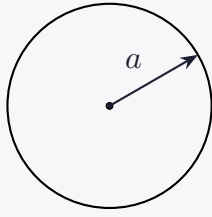
In free space (air-filled waveguide), $v_0 = c$. Hence, we prove the fundamental relation:

$$v_p \cdot v_g = c^2 \quad (\text{Q.E.D.}) \quad (7)$$

3. Circular Waveguide

A circular waveguide is a hollow tube of circular cross-section. Solving the wave equations in cylindrical coordinates (r, ϕ, z) requires Bessel functions.

Circular Waveguide Cross-Section



Cutoff Frequency Formula:

$$f_c = \frac{p'_{mn}c}{2\pi a} \quad (\text{for TE modes})$$

$$f_c = \frac{p_{mn}c}{2\pi a} \quad (\text{for TM modes})$$

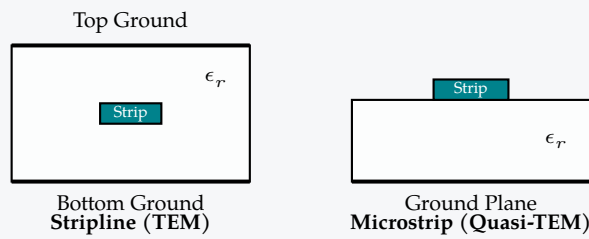
Dominant Mode is TE_{11} , where

$$p'_{11} \approx 1.8412.$$

4. Stripline and Microstrip Line

Striplines and microstrip lines are planar transmission lines fabricated on dielectric substrates, forming the core of Microwave Integrated Circuits (MICs).

Planar Line Geometries (Stripline vs. Microstrip)



4.1 Microstrip Design Formulas

Because some electric field lines pass through air and others through the substrate, the wave propagation is **Quasi-TEM**. We define an **effective dielectric constant** (ϵ_{eff}):

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{w} \right)^{-1/2} \quad (8)$$

Where w is the strip width and h is substrate thickness. The characteristic impedance is:

$$Z_0 = \frac{60}{\sqrt{\epsilon_{\text{eff}}}} \ln \left(8 \frac{h}{w} + \frac{w}{4h} \right) \quad (\text{for } w/h \leq 1) \quad (9)$$

Quick Revision — Module III

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Key Formulas at a Glance

Coaxial impedance	$Z_0 = \frac{138}{\sqrt{\epsilon_r}} \log_{10}(b/a)$
Rectangular cutoff	$f_c = \frac{c}{2\sqrt{\epsilon_r}} \sqrt{(m/a)^2 + (n/b)^2}$
Guide wavelength	$\lambda_g = \frac{\lambda_0}{\sqrt{1-(f_c/f)^2}}$
Velocity relation	$v_p \cdot v_g = c^2$ (in air-filled guide)
Circular cutoff	$f_c = \frac{p'_{mn}c}{2\pi a}$ (TE ₁₁ dominant, $p'_{11} = 1.8412$)
Effective dielectric	$\epsilon_{\text{eff}} \approx \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} (1 + 12h/w)^{-1/2}$

One-Line Definitions

- **Guide Wavelength:** The axial wavelength of a wave propagating inside a waveguide.
- **Dominant Mode:** The propagation mode with the absolute lowest cutoff frequency.
- **Phase Velocity:** The rate at which the wave's constant-phase wavefronts propagate.
- **Group Velocity:** The speed at which energy and information travel down the line.
- **Quasi-TEM Mode:** A mode resembling TEM but containing small longitudinal fields due to dielectric boundaries.
- **Stripline:** A planar transmission line sandwiched symmetrically between two ground planes.
- **Microstrip:** A planar line on a dielectric substrate with a single bottom ground plane.

Frequently Asked / Viva Questions

Q1. What is the dominant mode in rectangular waveguides and why?

The TE₁₀ mode. It has the lowest cutoff frequency ($f_c = c/2a$), meaning it can propagate over the widest single-mode bandwidth.

Q2. Why does a rectangular waveguide act as a high-pass filter?

Because it only supports wave propagation above its cutoff frequency ($f > f_c$). Waves below cutoff are attenuated exponentially.

Q3. Explain the relation between phase velocity, group velocity, and the speed of light.

In an air-filled guide, $v_p \cdot v_g = c^2$. Since $v_g < c$, the phase velocity v_p must be greater than the speed of light ($v_p > c$).

Q4. What is the dominant mode of a circular waveguide?

Frequently Asked / Viva Questions (contd.)

The TE_{11} mode, with a cutoff parameter $p'_{11} \approx 1.8412$.

Q5. Why are microstrip lines called Quasi-TEM lines?

Because the fields exist in two different mediums (air above, substrate below). The phase velocity mismatch prevents a pure TEM mode.

Q6. How does the characteristic impedance of a coaxial cable depend on conductor sizes?

It is proportional to the logarithm of the ratio of the outer radius to inner radius ($Z_0 \propto \ln(b/a)$).

Q7. Why are rectangular waveguides typically designed with $a = 2b$?

To maximize the frequency range over which only the dominant TE_{10} mode propagates before the next mode (TE_{20} or TE_{01}) turns on.

Q8. What is the physical meaning of the effective dielectric constant (ϵ_{eff})?

It is the equivalent permittivity of a uniform medium that would produce the same phase velocity as the air-substrate microstrip system.

Q9. What are the advantages of Striplines over Microstrip lines?

Striplines are shielded on both sides, which reduces radiation losses and provides better isolation against electromagnetic interference.

Q10. What is the cutoff wavelength (λ_c) of the dominant mode in a rectangular waveguide?

$\lambda_c = 2a$ for the TE_{10} mode.

Q11. What is a mode chart?

A plot of cutoff frequencies for various modes versus waveguide dimensions, used to select optimal waveguide sizes.

Q12. Under what condition does dispersion occur in transmission lines?

Dispersion occurs when the phase velocity of the propagating wave depends on frequency ($dv_p/df \neq 0$), causing signals of different frequencies to spread in time.

Comparison of Microwave Transmission Lines

Parameter	Coaxial Line	Rectangular Waveguide	Microstrip Line
Mode Supported	TEM	TE, TM	Quasi-TEM (TE/TM at high f)
Cutoff Freq f_c	0 (propagates DC)	$f_c = c/2a > 0$	0 (propagates DC)
Bandwidth	Very wide	Moderate (between modes)	Very wide
Losses	Moderate	Very low (no dielectric)	High (conductor + radiation)
Power Capacity	Moderate	High	Low